DTS103TC
Data Structure and Algorithms

Lab 3: Elementary Data Structures

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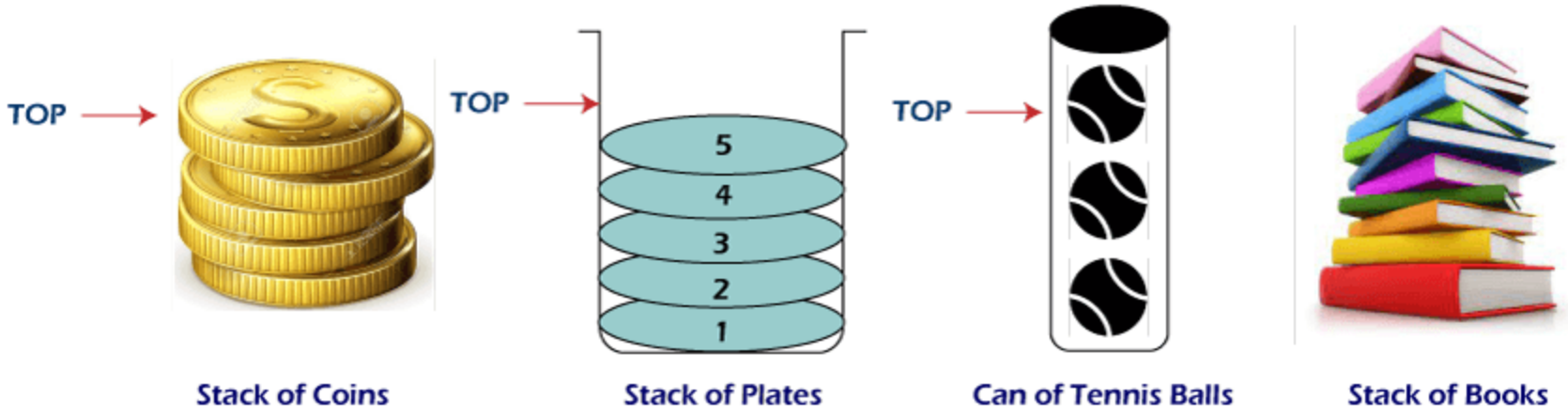
Learning outcomes

- Implementation of Data Structures
  - Stack
  - Queue
  - LinkedList
  - Heap and Heapsort

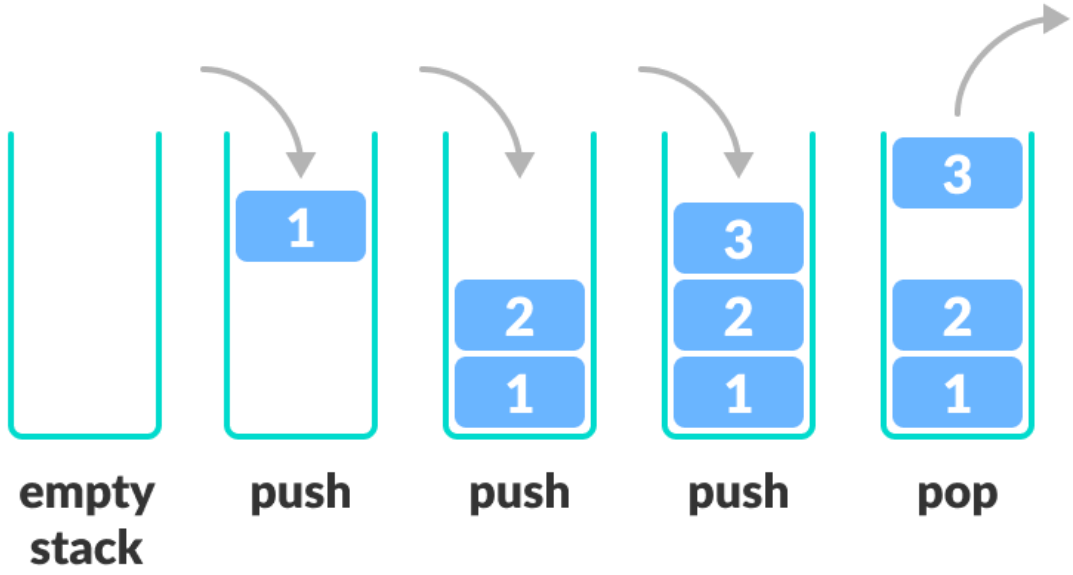
Stack

- A stack is a linear data structure that follows the principle of **Last In First Out (LIFO).**
  - the last element inserted inside the stack is removed first.



Basic Operations of Stack

- **Push:** Add an element to the top of a stack
- **Pop:** Remove an element from the top of a stack
- **Peek:** Get the value of the top element without removing it
- **IsEmpty:** Check if the stack is empty



Stack implementation – Python

- Python lists provide built-in methods that make them suitable for stack operations.

```
class Stack:

    def __init__(self):
        self.stack = []

    # check empty
    def isEmpty(self):
        return len(self.stack) == 0

    # Adding items into the stack
    def push(self, item):
        self.stack.append(item)
        print("pushed item: " + item)

    # Removing an element from the stack
    def pop(self):
        if (self.isEmpty()):
            return "underflow"
        return self.stack.pop()

    # Display the stack
    def display(self):
        print(self.stack)
```

Stack implementation – Python

```
stack = Stack()
stack.push(str(1))
stack.push(str(2))
stack.push(str(3))
stack.push(str(4))
print("popped item: " + stack.pop())
print("stack after popping an element: ")
stack.display()
```

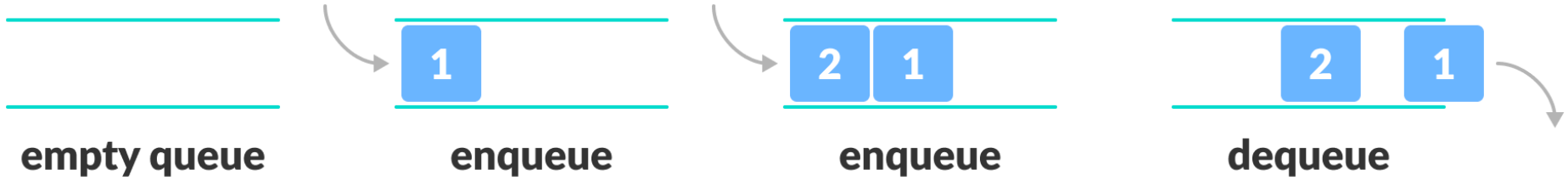
```
pushed item: 1
pushed item: 2
pushed item: 3
pushed item: 4
popped item: 4
stack after popping an element:
['1', '2', '3']
```

Queue

- Queue follows the First In First Out (FIFO) rule
  - the item that goes in first is the item that comes out first.

Basic Operations of Queue

- Enqueue: Add an element to the end of the queue
- Dequeue: Remove an element from the front of the queue
- IsEmpty: Check if the queue is empty
- Peek: Get the value of the front of the queue without removing it



Queue implementation – Python

- Python lists provide built-in methods that make them suitable for queue operations.

```
class Queue:

    def __init__(self):
        self.queue = []

    # Add an element
    def enqueue(self, item):
        self.queue.append(item)

    # Remove an element
    def dequeue(self):
        if (self.isEmpty()):
            return "underflow"
        return self.queue.pop(0)
```

```
def isEmpty(self):
        return len(self.queue) == 0

    # Display the queue
    def display(self):
        print(self.queue)
```

list.pop(0) is O(n)
- Must shift n-1 elements after removing the first element

Python collections.deque supports O(1) operations for inserting and removing elements at both ends

Queue Implementation - Python

```
q = Queue()
q.enqueue(1)
q.enqueue(2)
q.enqueue(3)
q.enqueue(4)
q.display()
q.dequeue()
print("After removing an element")
q.display()
```

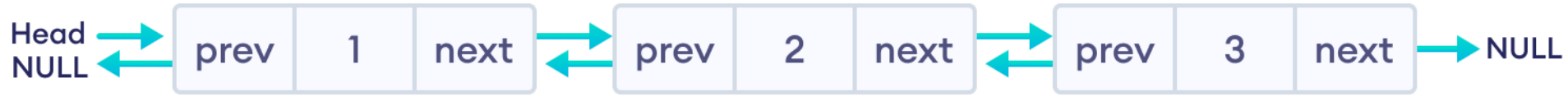
```
[1, 2, 3, 4]
After removing an element
[2, 3, 4]
```

Linked List

- A linked list is a linear data structure that includes a series of connected nodes.
- Unlike an array, the order in a linked list is determined by a pointer in each object.
- Linked lists can be of multiple types: singly, doubly, and circular linked list.

Doubly Linked List

- A doubly linked list is a type of linked list in which each node consists of 3 components:
  - prev - address of the previous node
  - data - data item
  - next - address of next node



- head points to the first node of the linked list.
- next pointer of the last node is NULL.

Operations of Linked List

- Traversal - access each element of the linked list
- Insertion - adds a new element to the linked list
- Deletion - removes the existing elements
- Search - find a node in the linked list

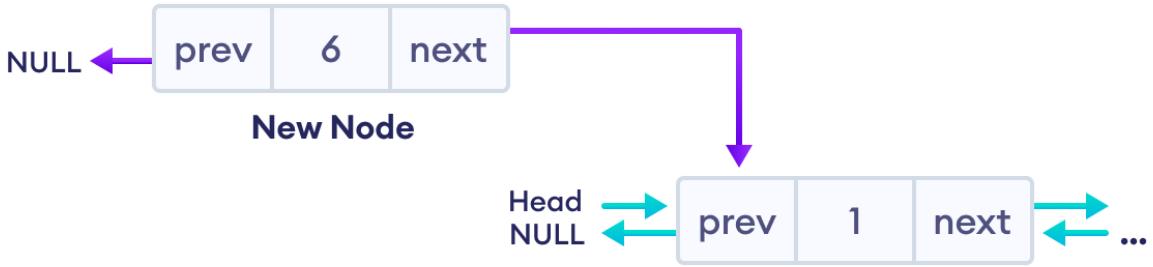
Insertion on a Doubly Linked List

- We can insert elements at 3 different positions of a doubly-linked list:
  - Insertion at the beginning
  - Insertion in-between nodes
  - Insertion at the End

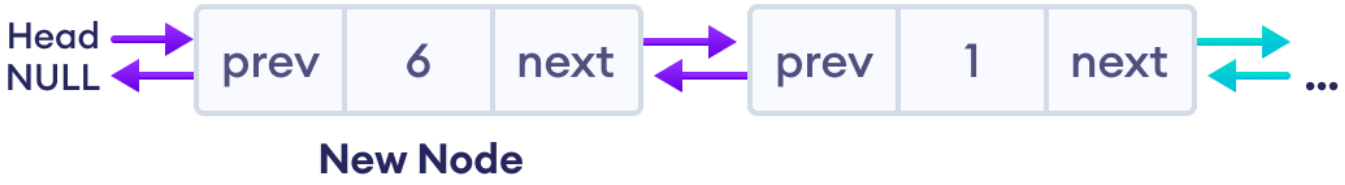
Insertion at the Beginning

- 1. Create a new node

- 2. Set prev and next pointers of new node



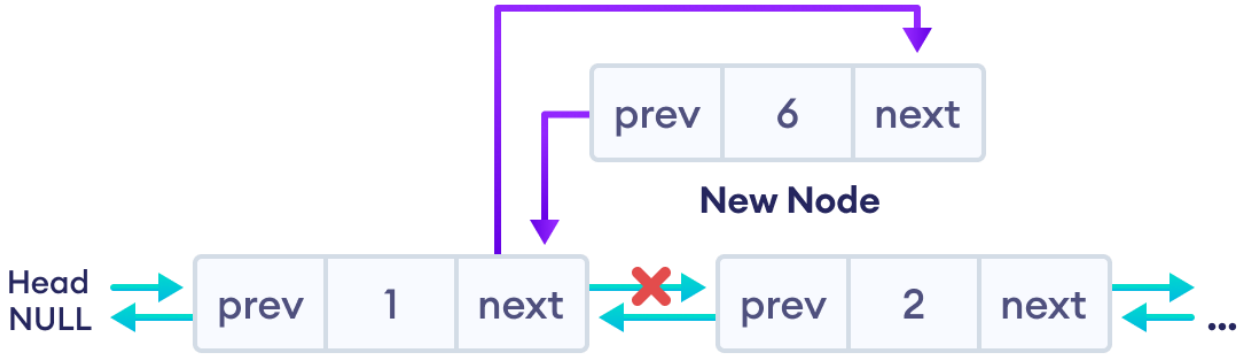
- 3. Make new node as head node



Insertion in between two nodes

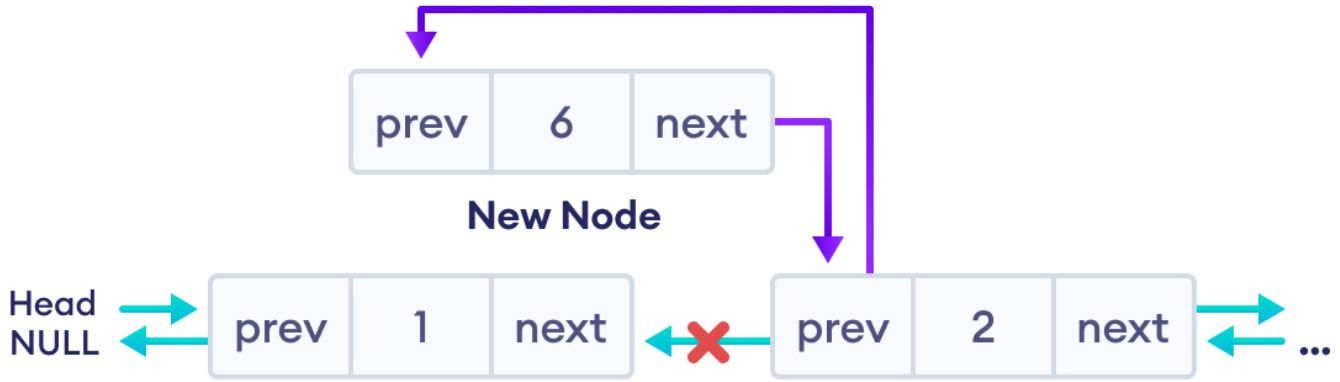
- 1. Create a new node

- 2. Set the next pointer of new node and previous node



Insertion in between two nodes

- 3. Set the prev pointer of new node and the next node

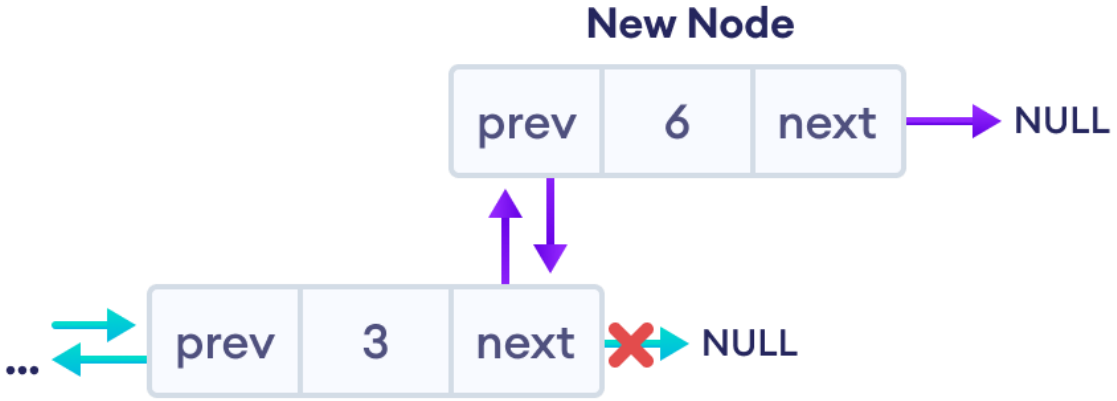


- The final doubly linked list is after this insertion is:

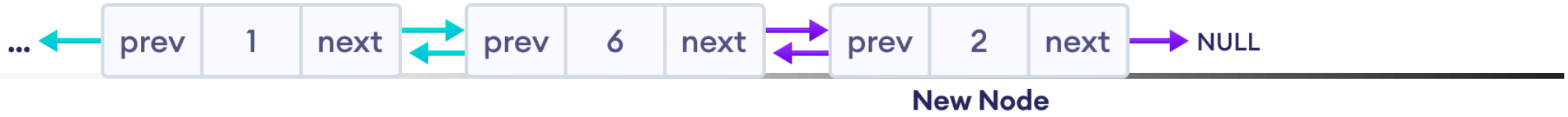
Insertion at the End

- 1. Create a new node

- 2. Set prev and next pointers of new node and the previous node



- The final doubly linked list looks like this.

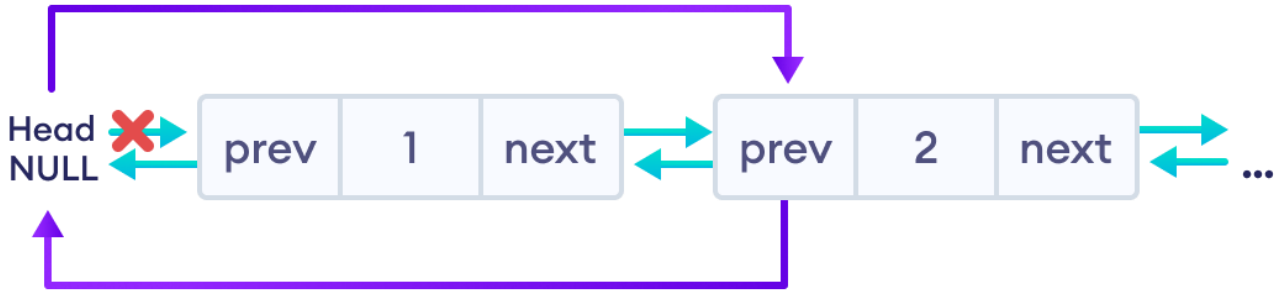


Deletion from a Doubly Linked List

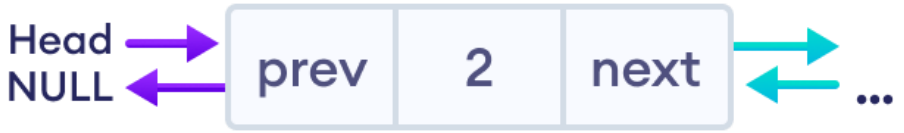
- Similar to insertion, we can also delete a node from 3 different positions of a doubly linked list.
  - 1. Delete the First Node of Doubly Linked List
  - 2. Deletion of the Inner Node
  - 3. Delete the Last Node of Doubly Linked List

Delete the First Node of Doubly Linked List

- If the node to be deleted (i.e. del_node) is at the beginning.
  - Reset value node after the del_node (i.e. node two)



  - Free the memory of del_node, and the linked list will look like this.

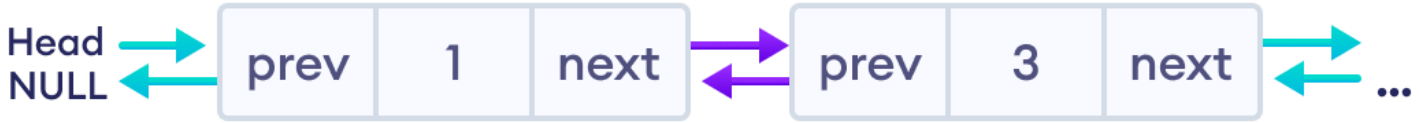


Free the space of the first node

Deletion of the Inner Node

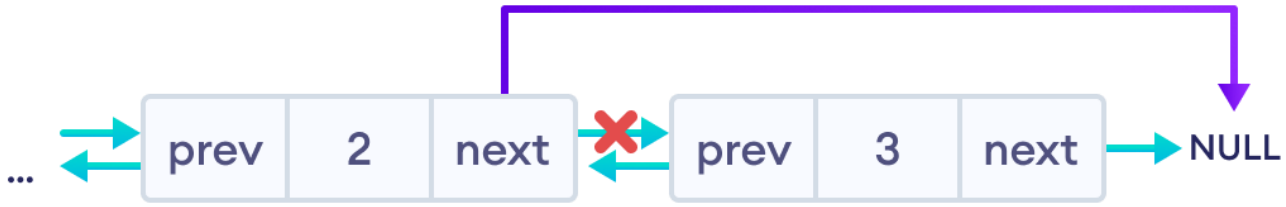
- If del_node is an inner node
  - reset the value of next and prev of the nodes before and after the del_node.

  - Free the memory of del_node, and the linked list will look like this.

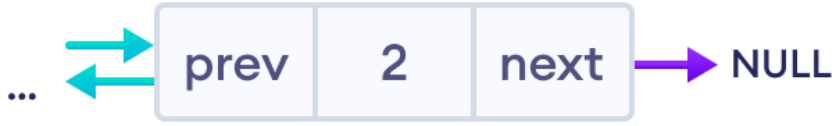


Delete the Last Node of Doubly Linked List

- If the node to be deleted (i.e. del_node) is at the End.
  - simply delete the del_node and make the next of node before del_node point to NULL.



  - Free the memory of del_node, and the linked list will look like this.



Lab Exercises

- Implementation of Data Structures
  - Stack
  - Queue
  - LinkedList
  - Heap and Heapsort (Refer to Lecture Slides)